

A spherical-cap obstruction for Gaussian equal-mass centroid partitions

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This is a continuation of the mass-constrained Gaussian partition programme and the previously released dimension-rate theorem. All inequalities below concern **arbitrary measurable partitions**, not only conical fans, power diagrams or the explicit cyclic constructions.

Main results

Let γ_d denote the standard Gaussian law on $\mathbb{R}^d$, and

$$F_d(k) = \sup_{\substack{(A_1, \dots, A_k) \text{ partition} \\ \gamma_d(A_i) = 1/k}} \sum_{i=1}^k \left\| \int_{A_i} x d\gamma_d(x) \right\|^2.$$

Set $L = \log k$, $h_k = k\varphi(\Phi^{-1}(1 - 1/k))$, $U_k = h_k^2/k$, and $F_\infty(k) = F_{k-1}(k)$. Our prior exact all-mass theorem gives

$$0 \leq U_k - F_\infty(k) \leq 2/k. \quad (1)$$

Theorem A (quantitative finite-dimensional spherical-cap converse). For $k \geq 2, d \geq 3$, and any $s \in (0, \sqrt{d})$ satisfying

$$\mathcal{B}_{d,k}(s) := \frac{k}{s\sqrt{2\pi(1-1/d)}} \left(1 - \frac{s^2}{d}\right)^{(d-1)/2} \leq 1, \quad (2)$$

the global partition optimum satisfies the exact nonasymptotic bound

$$\boxed{kF_d(k) \leq h_k \left(s + \frac{d}{(d-1)s} \right)}. \quad (3)$$

No orientation, regularity, symmetry, or assumed optimizer enters.

Theorem B (explicit quadratic-logarithmic dimensional defect). For every integer k with $L = \log k \geq 100$, and every integer $d \geq L^2/\log L$,

$$\boxed{U_k - F_d(k) \geq \frac{1}{k} \left(\frac{L^2}{d} - 14 \right)}. \quad (4)$$

The right side may be negative for large d , in which case (4) remains correct but is not informative.

Theorem C (quadratic-logarithmic necessary dimension). Fix a finite constant $C \geq 0$. If for all sufficiently large k there exists an exactly equal-mass Gaussian k -partition in dimension d_k with

$$P(\mathcal{A}_k) \geq F_\infty(k) - C/k,$$

then

$$\liminf_{k \rightarrow \infty} \frac{d_k}{(\log k)^2} \geq \frac{1}{C + 16}. \quad (5)$$

Therefore **bounded additive $O(1/k)$ error necessarily requires dimension $\Omega((\log k)^2)$** .

The preceding cubic-logarithmic *sufficient* dimension bound, $d \leq 8(\log k)^3 + 2$ for additive $14/k$ error, is unchanged. The remaining gap is now exactly between quadratic and cubic logarithms, not between a quadratic-logarithmic / loglog lower and cubic-logarithmic upper.

1. From any partition to a Gaussian spherical code

Lemma 1 (centroid directions turn partitions into maxima). For any k -partition of exactly equal Gaussian masses, let $b_i = \int_{A_i} x d\gamma_d(x)$ and $u_i \in S^{d-1}$ be any unit direction along b_i (arbitrary when $b_i = 0$). Then

$$\frac{k}{h_k} \sum_i \|b_i\|^2 \leq \mathbb{E} \max_i \langle u_i, G \rangle, \quad G \sim N(0, I_d). \quad (6)$$

Proof. For every measurable A of Gaussian mass $1/k$, the sharp one-cell halfspace rearrangement gives

$$\left\| \int_A x d\gamma_d(x) \right\| \leq \varphi(\Phi^{-1}(1 - 1/k)) = h_k/k.$$

Indeed the pointwise comparison of the set with the Gaussian upper halfspace having the same mass and normal direction as its moment has a nonnegative signed threshold integral. Consequently, $\sum_i \|b_i\|^2 \leq (h_k/k) \sum_i \|b_i\|$. For each x , $\max_i \langle u_i, x \rangle \geq \sum_i \mathbf{1}_{A_i}(x) \langle u_i, x \rangle$. Integrating gives

$$\mathbb{E} \max_i \langle u_i, G \rangle \geq \sum_i \langle u_i, b_i \rangle = \sum_i \|b_i\|.$$

Combine the two inequalities. $\square$

This is the central structural bridge: it is valid even for extremely irregular partitions and converts their first-moment score into a maximum of only k Gaussian linear forms.

2. A uniform spherical-cap tail

Let V be uniform on S^{d-1} for $d \geq 3$, and $R = \|G\|$. The standard Gaussian polar decomposition $G = RV$ makes R, V independent with $\mathbb{E}R \leq \sqrt{d}$.

Lemma 2 (a cap estimate with its polynomial prefactor). For $d \geq 3$ and every $0 < a < 1$,

$$\Pr(V_1 \geq a) \leq \frac{1}{a\sqrt{2\pi(d-1)}} (1 - a^2)^{(d-1)/2}. \quad (7)$$

Proof. The first spherical coordinate has density

$$c_d(1 - u^2)^{(d-3)/2} \mathbf{1}_{\{|u| < 1\}}, \quad c_d = \frac{\Gamma(d/2)}{\sqrt{\pi} \Gamma((d-1)/2)}.$$

Log-convexity of the Gamma integral (by Hölder's inequality) gives $\Gamma(z+1/2)^2 \leq \Gamma(z)\Gamma(z+1) = z\Gamma(z)^2$ for $z > 0$. Hence $c_d \leq \sqrt{(d-1)/(2\pi)}$. Since $u \geq a$,

$$\begin{aligned} \int_a^1 (1-u^2)^{(d-3)/2} du &\leq \frac{1}{a} \int_a^1 u(1-u^2)^{(d-3)/2} du \\ &= \frac{(1-a^2)^{(d-1)/2}}{(d-1)a}. \end{aligned}$$

Multiplying the estimates proves (7). $\square$

Lemma 3 (integrating a spherical cap union). For any k unit directions in $\mathbb{R}^d$, define

$$B_{d,k}(a) = \frac{k}{a\sqrt{2\pi(d-1)}}(1-a^2)^{(d-1)/2}.$$

If $B_{d,k}(a) \leq 1$, then

$$\boxed{\mathbb{E} \max_{1 \leq i \leq k} \langle u_i, G \rangle \leq \sqrt{d} \left(a + \frac{1}{(d-1)a} \right)}. \quad (8)$$

Proof. Put $T = \max_i \langle u_i, V \rangle$. The union bound and Lemma 2 give $\Pr(T \geq u) \leq B_{d,k}(u)$ for $a \leq u < 1$. Direct differentiation shows

$$\frac{d}{du} \log B_{d,k}(u) = -\frac{1}{u} - \frac{(d-1)u}{1-u^2} \leq -(d-1)a \quad (u \geq a).$$

Therefore $B_{d,k}(u) \leq B_{d,k}(a) \exp[-(d-1)a(u-a)]$, and hence

$$\mathbb{E} T \leq a + \mathbb{E}(T-a)_+ \leq a + \int_a^1 B_{d,k}(u) du \leq a + \frac{B_{d,k}(a)}{(d-1)a} \leq a + \frac{1}{(d-1)a}.$$

Independence and $\mathbb{E} R \leq \sqrt{d}$ prove (8). There is no implicit assumption that the maximum is positive: the first inequality $T \leq a + (T-a)_+$ holds pointwise. $\square$

Proof of Theorem A. Choose $a = s/\sqrt{d}$ in Lemma 3. Then $B_{d,k}(a) = \mathcal{B}_{d,k}(s)$, and (8) becomes

$$\mathbb{E} \max_i \langle u_i, G \rangle \leq s + \frac{d}{(d-1)s}.$$

Insert this into Lemma 1 and take the supremum over partitions. $\square$

3. A two-log-scale cap threshold

Lemma 4 (universal explicit threshold). Suppose $L = \log k \geq 100$, $d \geq L^2/\log L$, and set

$$S = 2L - \log L - \frac{2L^2}{d} + 16, \quad s = \sqrt{S}. \quad (9)$$

Then $0 < S < d$, $\mathcal{B}_{d,k}(s) < 1$, and

$$\boxed{\left(s + \frac{d}{(d-1)s} \right)^2 \leq 2L - \log L - \frac{2L^2}{d} + 24.} \quad (10)$$

Proof. The elementary inequality $\log L \leq \sqrt{L}$ holds for $L \geq 1$. In particular, for $L \geq 100$,

$$d \geq L^2/\log L \geq L^{3/2} \geq 10L.$$

Also $L^2/d \leq \log L$, so

$$S \geq 2L - 3\log L + 16 \geq L, \quad S \leq 2L + 16 \leq 3L < d.$$

The logarithm of (2), using $\log(1-x) \leq -x - x^2/2$ on $0 < x < 1$, satisfies

$$\begin{aligned} \log \mathcal{B}_{d,k}(s) &\leq L - \frac{1}{2} \log S - \frac{1}{2} \log(2\pi(1-1/d)) \\ &\quad - \frac{S}{2} - \frac{S^2}{4d} + \frac{S}{2d} + \frac{S^2}{4d^2}. \end{aligned} \tag{11}$$

Write $\Delta = 2L - S = \log L + 2L^2/d - 16$. A direct expansion gives

$$L - \frac{S}{2} - \frac{S^2}{4d} = \frac{1}{2} \log L - 8 + \frac{L\Delta}{d} - \frac{\Delta^2}{4d}.$$

Here $\Delta \leq 3\log L$, and hence $L\Delta/d \leq 3(\log L)^2/L \leq 3$. Moreover $S \geq L$, $S/(2d) \leq 3/20$, $S^2/(4d^2) \leq 9/400$, and $2\pi(1-1/d) > 1$. Inserting these inequalities into (11) yields

$$\log \mathcal{B}_{d,k}(s) \leq -8 + 3 + \frac{3}{20} + \frac{9}{400} < 0.$$

Thus the cap union is strictly below one.

Since $d \geq 3$, $d/(d-1) \leq 2$, and $S \geq 1$,

$$\left(s + \frac{d}{(d-1)s}\right)^2 \leq (s + 2/s)^2 = S + 4 + 4/S \leq S + 8.$$

Substitution of S proves (10). $\square$

Lemma 5 (lower normal-tail centroid quantile). For $k \geq e^{100}$ with $L = \log k$,

$$\boxed{h_k^2 \geq 2L - \log L - 4.} \tag{12}$$

Proof. Put $t = \Phi^{-1}(1-1/k) > 1$. The elementary Mills bounds $t \leq h_k \leq t + 1/t$ and the lower Mills bound $\bar{\Phi}(t) \geq t\varphi(t)/(1+t^2)$ yield

$$L \leq \frac{t^2}{2} + \log(t + 1/t) + \frac{1}{2} \log(2\pi).$$

The upper Mills bound gives $t^2 \leq 2L$. For $t \geq 1$, $t + 1/t \leq 2t \leq 2\sqrt{2L}$, and therefore

$$t^2 \geq 2L - \log L - \log(16\pi) > 2L - \log L - 4.$$

The final elementary strict inequality follows, for instance, from $\pi < 22/7$ and $e > 8/3$: $16\pi < 352/7 < 4096/81 < (e)^4$. Since $h_k \geq t$, (12) follows. $\square$

Proof of Theorem B. Combine Lemmas 1, 4 and Theorem A to obtain

$$kF_d(k) \leq h_k H, \quad H^2 \leq 2L - \log L - \frac{2L^2}{d} + 24.$$

By Lemma 5, $h_k^2 - H^2 \geq 2L^2/d - 28$. The elementary identity

$$h_k(h_k - H) - \frac{h_k^2 - H^2}{2} = \frac{(h_k - H)^2}{2} \geq 0$$

holds for all $h_k, H \geq 0$. Thus

$$h_k^2 - kF_d(k) \geq h_k^2 - h_k H \geq \frac{h_k^2 - H^2}{2} \geq \frac{L^2}{d} - 14.$$

Divide by k , proving (4). $\square$

4. Consequences for the exact dimension problem

Proof of Theorem C. Write $L = \log k$ and fix $C < \infty$. The inherited universal equal-mass construction (1) implies

$$kF_\infty(k) \geq h_k^2 - 2.$$

Therefore the assumed partition obeys $kF_{d_k}(k) \geq h_k^2 - C - 2$.

Set $D_0 = \lceil L^2 / \log L \rceil$. If $d_k < D_0$, lift its partition cylindrically to dimension D_0 , without changing the objective. Theorem B applied in dimension D_0 then implies

$$\frac{L^2}{D_0} \leq C + 16.$$

But $L^2/D_0 \sim \log L \rightarrow \infty$, a contradiction for all sufficiently large k . Hence $d_k \geq D_0$. Applying Theorem B at the actual dimension d_k gives

$$h_k^2 - C - 2 \leq kF_{d_k}(k) \leq h_k^2 - \frac{L^2}{d_k} + 14,$$

and therefore

$$d_k \geq \frac{L^2}{C + 16}.$$

This holds for all sufficiently large k , proving (5). $\square$

Comparison with the prior result. The earlier entropy / Gaussian rate–distortion converse established only $d \geq (2 - o(1))L^2 / \log L$ for additive $O(1/k)$ accuracy. Our sphere-union argument retains the $-\log L$ correction to the normal extreme tail **and** the finite-dimensional L^2/d deficit simultaneously. That yields the stronger order L^2 , not just a constant improvement to the earlier argument.

The upper construction $d \leq 8L^3 + 2$ is inherited from the cyclic-orbit Gaussian partition theorem. No $O(L^2)$ construction is established here, so a remaining logarithmic-factor gap persists. This note does not claim to settle the full exact dimension-rate function or the fixed- k Standard Simplex optimality question.

5. Sources, novelty and verification contract

Mathematical inputs cited (and, where used critically, reproved): the one-cell Gaussian rearrangement, Mills inequalities, Gaussian polar independence, gamma log-convexity, and exact mass-envelope $U_k - F_\infty(k) \leq 2/k$. The first four are classical; the last is proved in the companion research note.

The classical spherical-cap union bound and its connection to Gaussian process maxima have broad antecedents in spherical code and high-dimensional geometry literature. We do not claim that this exact form is historically new without a wider specialist survey.

The checker supplies rational interval evidence of the threshold algebra for several integer k, d pairs without using floating-point certification. In particular the universal lemma is established analytically, not by numerical enumeration. All proof dependencies and scope restrictions are disclosed in AUDIT.md.